

Central Charges: Susy $\{Q_\alpha, Q_\beta\} = P_\mu \delta_{\alpha\beta}^\mu$

* Even dim: $\{Q_\alpha, Q_\beta\} = (P_\mu \gamma_\mu C^{-1})_{\alpha\beta} P^\mu + (P_\mu \gamma_{\mu_1 \dots \mu_r} C^{-1})_{\alpha\beta} Z^{\mu_1 \dots \mu_r}$

$\rightarrow$ # of Central charges is $16 + \frac{1}{2} \binom{16}{5,1 \times 5,1} = 136$
 $D=3 \pmod{4}$, γ_{D-2} behaves exactly like γ under complex conj. and transpose

$$\left(\begin{array}{c} \gamma_{\mu}, \mu=0,1,\dots,D-2 \\ (C \gamma_{\mu_1 \dots \mu_r})^T = (-1)^{(D-1)(r-1)/2} (C \gamma_{\mu_1 \dots \mu_r}) \end{array} \right)$$

$$\mu_1, \dots, \mu_r = 0, \dots, D-1$$

Proof: C is unitary: $C^\dagger C = \mathbb{1}$

s.t. $C \Gamma^A$ is symm or anti symm

$$(C \Gamma^{(\mu)})^T = -t_r C \Gamma^{(\mu)}, \quad t_r = \pm 1$$

$$\left(\begin{array}{c} \Gamma^A = \mathbb{1}, \gamma^\mu, \dots, \gamma^{\mu_1 \dots \mu_r} \\ \mu_1 < \mu_2 < \dots < \mu_r \end{array} \right)$$

$$\underline{r=0}: (C \mathbb{1})^T = -t_r C \mathbb{1}$$

$$C^T = -t_0 C$$

$$\underline{r \geq 1}: (C \gamma^\mu)^T = -t_1 C \gamma^\mu$$

$$(\gamma^\mu)^T C^T = -t_1 C \gamma^\mu$$

$$(\gamma^\mu)^T (-t_0 C) = -t_1 C \gamma^\mu$$

$$(\gamma^\mu)^T = (t_0 t_1) C \gamma^\mu C^{-1}$$

t_0, t_1 : depend on $D \pmod{8}$ & rank $r \pmod{4}$

$$\psi_m(x) \rightarrow \psi_m(x) + \partial_\mu \epsilon(x)$$

$$\text{Global SUSY} \Rightarrow D \epsilon(x) = 0$$

ψ_{ma} transforms in the reps of $\text{Spin}(D-1) \times \text{SO}(D-1)$

$\psi_{ma} \in \mathbb{C} \otimes \text{Spin}(D-1)$ with rank $2^{\lfloor D/2 \rfloor}$

Gravitino Action: $S = - \int d^D x \bar{\psi}_m \gamma^{mnp} \partial_n \psi_p$

* Action is gauge invariant but $\mathcal{L}$ is not.
the variation of $\mathcal{L}$ is invariant up to a total deriv.

$$\delta \mathcal{L} = - \partial_n (\bar{\epsilon} \gamma^{mnp} \partial_n \psi_p)$$

E.O.M: $S = - \int d^D x \bar{\psi}_m \gamma^{mnp} \partial_n \psi_p$

ψ_m : Dirac

$$\partial_m \frac{\delta \mathcal{L}}{\delta (\partial_m \psi)} - \frac{\delta \mathcal{L}}{\delta \psi} = 0$$

$$\gamma^{mnp} \partial_n \psi_p = 0 \quad (\text{Gravitino E.O.M})$$

is similar to $\partial^m F_{mn} = 0$

Gauge invariance: $\psi_p \rightarrow \psi_p + \partial_p \epsilon$

$$\begin{aligned} \gamma^{mnp} \partial_n \psi_p &\rightarrow \gamma^{mnp} \partial_n (\psi_p + \partial_p \epsilon) \\ &= \gamma^{mnp} \partial_n \psi_p + \gamma^{mnp} \partial_n \partial_p \epsilon = 0 \end{aligned}$$

where $\gamma^{mnp} \partial_n \partial_p \epsilon = 0$

• likewise

$$\partial_m \gamma^{mnp} \partial_n \psi_p = 0$$

• Thus, we have two ^{less} symmetries which

constrain the components of ψ . So,
a Gravitino has

* ψ : $2^{\lfloor D/2 \rfloor}$ components
 $\mu = 0, \dots, D-1$

which yields

$$(D-1) 2^{\lfloor D/2 \rfloor}$$

possible equations for $\gamma^{\mu\nu\rho} \partial_\nu \psi_\rho = 0$. As such
we can determine $D 2^{\lfloor D/2 \rfloor}$ components of ψ
up to Gauge freedom

Off-shell D.O.F.: $(D-1) 2^{\lfloor D/2 \rfloor}$

$$\underline{D=3}: \gamma^{\mu\nu\rho} \partial_\nu \psi_\rho = 0$$

$$\gamma^{\mu\nu\rho} = \alpha \epsilon^{\mu\nu\rho}$$

$$\Rightarrow \alpha \epsilon^{\mu\nu\rho} \partial_\nu \psi_\rho = 0$$

$$\epsilon_{\mu_1 \dots \mu_n \nu_1 \dots \nu_p} \epsilon^{\mu_1 \dots \mu_n \sigma_1 \dots \sigma_p} = \alpha! \delta_{\nu_1 \dots \nu_p}^{\sigma_1 \dots \sigma_p}$$

$$\begin{aligned} \epsilon_{\lambda\sigma\mu} \epsilon^{\mu\nu\rho} \partial_\nu \psi_\rho &= 0 \\ &= -\delta^{\nu\rho}_{\sigma\mu} \partial_\nu \psi_\rho = -\left(\frac{1}{2!}\right) \delta^\nu_\sigma \delta^\rho_\mu \partial_\nu \psi_\rho = 0 \\ &= -\frac{1}{2} (\delta^\nu_\sigma \delta^\rho_\mu - \delta^\rho_\sigma \delta^\nu_\mu) \partial_\nu \psi_\rho \\ &= -\frac{1}{2} (\partial_\sigma \psi_\mu - \partial_\mu \psi_\sigma) = 0 \end{aligned}$$

In $D=3$: $\partial_\sigma \psi_\mu - \partial_\mu \psi_\sigma = 0$
and so ψ_μ has no gauge invariant freedom

$$(\partial_\sigma \partial_\mu - \partial_\mu \partial_\sigma) \psi + (\dots) = 0$$

and so no propagating particle occur
in $D=3$.

$\hookrightarrow$ SUSY counterpart of $R_{\mu\nu} = 0$ in $D=3$

$$\star \quad \gamma_\mu \gamma^\mu = D, \quad \gamma'_\mu \gamma^{\mu\nu} = (D-1) \gamma^\nu$$

$$\gamma_\mu \gamma^{\mu\nu\rho} = (D-2) \gamma^{\nu\rho}$$

$\Rightarrow$

$$\gamma^{\mu\nu\rho} \partial_\nu \psi_\rho = 0 \Rightarrow \gamma_\mu \gamma^{\mu\nu\rho} \partial_\nu \psi_\rho = 0$$

$$= (D-2) \gamma^{\nu\rho} \partial_\nu \psi_\rho = 0$$

or $\gamma^{\nu\rho} \partial_\nu \psi_\rho = 0$ in $D > 2$

$$\gamma^{\mu\nu\rho} = \gamma^\mu \gamma^{\nu\rho} = 2 \eta^{\mu[\nu} \gamma^{\rho]}$$

$$\Rightarrow \gamma_\mu \gamma^{\mu\nu\rho} = D \gamma^{\nu\rho} - 2 \gamma_\mu \eta^{\mu[\nu} \gamma^{\rho]}$$

$$\gamma^{\nu\rho} \partial_\nu \psi_\rho (D-2) = D \gamma^{\nu\rho} \partial_\nu \psi_\rho - 2 \gamma_\mu \eta^{\mu[\nu} \gamma^{\rho]} \partial_\nu \psi_\rho$$

$$\gamma^{\nu\rho} \partial_\nu \psi_\rho = \gamma_\mu \eta^{\mu[\nu} \gamma^{\rho]} \partial_\nu \psi_\rho$$

$$\gamma^{\nu\rho} \partial_\nu \psi_\rho = \gamma_\mu \frac{1}{2} [\eta^{\mu\nu} \gamma^\rho - \eta^{\mu\rho} \gamma^\nu] \partial_\nu \psi_\rho = 0$$

$$\gamma^\nu \gamma^\rho \partial_\nu \psi_\rho - \gamma^\rho \gamma^\nu \partial_\nu \psi_\rho = 0 \quad (\star)$$

$$\Rightarrow (\gamma^\mu \gamma^{\nu\rho} - 2 \eta^{\mu[\nu} \gamma^{\rho]}) \partial_\nu \psi_\rho = 0$$

$$\therefore \eta^{\mu[\nu} \gamma^{\rho]} \partial_\nu \psi_\rho = 0$$

$$\eta^{\mu\nu} \gamma^\rho - \eta^{\mu\rho} \gamma^\nu \partial_\nu \psi_\rho = 0$$

$$(\delta^\nu_\lambda \gamma^\rho - \delta^\rho_\lambda \gamma^\nu) \partial_\nu \psi_\rho = 0$$

$$\Rightarrow \gamma^\rho \partial_\lambda \psi_\rho - \gamma^\rho \partial_\rho \psi_\lambda = 0$$

$$\therefore \gamma^\rho (\partial_\lambda \psi_\rho - \partial_\rho \psi_\lambda) = 0$$